✅ 1. Competency #5: Applying mathematical concepts and rules in real-world contexts

- Highest relevance.
- Specific Analysis:
  - The task requires students to apply an abstract mathematical operation (percentage discount) to a concrete shopping scenario.
  - Students must understand the mathematical implication of the business rule: "20% off for two or more items" translates to multiplying the total by 0.8 .
  - Key Aspect: Students need to recognize the threshold ("buy one" vs. "buy two"). Although all options involve two or more items, understanding this rule is the prerequisite for correct calculation.
  - This embodies the core *interpret.pdf* principle of "translating real-world constraints into mathematical operations."

✅ 2. Competency #8: Evaluating the reasonableness and feasibility of results

- Highly relevant.
- Specific Analysis:
  - The task is not merely to calculate, but to judge (Yes/No). Students must compute the discounted total and compare it to Jason's budget (200 zeds).
  - This demands decision-making ability: Is the calculated result $\leq$ 200?
  - Contextual Evaluation: For example, for "All three items," the original sum is $155 + 86 + 79 = 320$. At 20% off, it becomes 256 . Students must evaluate that $256 > 200$, leading to the conclusion "No."
  - This judgment based on a budgetary limit is a classic example of *interpret.pdf*'s focus on "assessing the meaning of results within a real situation."

✅ 3. Competency #3: Interpreting mathematical information in texts, symbols and diagrams

- Highly relevant.
- Specific Analysis:
  - Information Extraction: Students need to extract key values (prices, 20% discount, 200 zeds) and conditions ("two or more items") from the text.
  - Symbolic Understanding: Understanding that "20% off" mathematically represents a coefficient of 0.8 .
  - Structural Mapping: The question presents options in a table. Students must map their calculated mathematical results back to the "Yes/No" choices in the table, completing the loop of information interpretation.

✅ 4. Competency #10: Dealing with uncertainty and estimation

- Moderately relevant.
- Specific Analysis:
  - While the task leans toward exact calculation, in practice, students might use estimation strategies to quickly assess viability.
  - For instance, for "MP3 Player + Headphones": The original price is roughly $155 + 90 = 245$. Taking off 20% (roughly 50) gives approximately 195 , which is close to 200. This forces students to consider whether to calculate exactly or verify after estimating.
  - This judgment at the edge of the budget reflects the *interpret.pdf* competency regarding "handling numerical boundaries and uncertainty."

✅ 5. Competency #11: Understanding how mathematical concepts vary across contexts

- Moderately relevant.
- Specific Analysis:
  - The discount mechanism (threshold-based) demonstrates a non-linear change in the mathematical model.
  - Although the task doesn't directly test single items, the context implies that the price structure changes drastically between "buying one" and "buying two." Students need to understand how the variable "quantity" alters the pricing function (from no discount to discount).
  - This reflects the *interpret.pdf* requirement to understand "how changing the context alters the mathematical relationship."

✅ 1. 能力 #5：在现实情境中应用数学规则和概念 (Applying mathematical concepts and rules in real-world contexts)

- 最高相关性。
- 具体分析：
  - 题目要求学生将抽象的数学运算（百分比折扣）应用于具体的购物情境。
  - 学生必须理解“购买两件或以上打八折”这一商业规则的数学含义：即总价格 $\times 0.8$ 。
  - 关键点：学生需要识别出“单独买一件不打折，买两件才打折”的临界点。虽然本题选项都是两件或以上，但理解这一规则是正确计算的前提。
  - 这体现了 *interpret.pdf* 中强调的“将现实世界的约束转化为数学运算”的核心能力。

✅ 2. 能力 #8：评估结果的合理性与可行性 (Evaluating the reasonableness and feasibility of results)

- 高度相关。
- 具体分析：
  - 任务不仅仅是计算，而是判断（Yes/No）。学生需要计算打折后的总价，并将其与 Jason 的预算（200 zeds）进行比较。
  - 这要求学生具备决策能力：计算结果是否 $\leq$ 200?
  - 情境评估：例如，对于“全部三件商品”，原价总和为 $155 + 86 + 79 = 320$ 。打八折后为 256 。学生必须评估 $256 > 200$ ，从而得出“买不起”的结论（No）。
  - 这种基于预算限制的“可行/不可行”判断，是 *interpret.pdf* 中关于“评估结果在现实情境中的意义”的典型体现。

✅ 3. 能力 #3：解释文本、符号和图表中的数学信息 (Interpreting mathematical information in texts, symbols and diagrams)

- 高度相关。
- 具体分析：
  - 信息提取：学生需要从题目描述中提取关键数值（价格、20%折扣、200 zeds预算）和条件（“two or more items”）。
  - 符号理解：理解“20% off”在数学上代表 $1 - 0.2 = 0.8$ 倍的系数。
  - 结构理解：题目最后以表格形式呈现三个选项，学生需要将计算出的数学结果映射回表格中的“Yes/No”选项，完成信息的闭环解释。

✅ 4. 能力 #10：处理不确定性与估算 (Dealing with uncertainty and estimation)

- 中等相关性。
- 具体分析：
  - 虽然题目鼓励精确计算，但在实际操作中，学生可能会使用估算策略来快速判断。
  - 例如，对于“MP3播放器+耳机”：原价约 $155 + 90 = 245$，打八折（约打两折即约50）约为 195，接近200。这迫使学生思考是否需要精确计算，或者在估算后进行验证。
  - 这种在预算边缘（接近200）的判断，体现了 *interpret.pdf* 中关于“处理数值边界和不确定性”的能力。

✅ 5. 能力 #11：理解数学概念在不同情境下的变化 (Understanding how mathematical concepts vary across contexts)

- 中等相关性。
- 具体分析：
  - 题目中的折扣机制（Threshold-based discount）展示了数学概念的非线性变化。
  - 虽然本题未直接考察单件商品，但情境暗示了“买一件”和“买两件”的价格结构完全不同。学生需要理解数量的变化如何改变定价模型（从无折扣到有折扣）。
  - 这反映了 *interpret.pdf* 中关于“理解情境变化如何影响数学关系”的要求。